

ADITYA S

MASTER RESUME • ALL MATERIAL • NOT FOR SUBMISSION — ASSEMBLE A VARIANT FROM THIS

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This is the source document. Every claim below is defensible. To target an unlisted role, copy this file, delete what the JD does not ask for, and keep it to one page. The five ready-made variants in this folder cover Software Engineering, AI/ML, Data Analytics, Product & Strategy, and QA.

EDUCATION

B.E., Information Science and Engineering — Atria Institute of Technology (VTU)

Expected May 2027

Bangalore, India

CGPA 8.22 / 10.0 through Semester VI • No Backlogs

Class XII (PUC) 86.16%, 2023 • Class X (CBSE) 90.0%, 2021

Semester SGPA: I 7.90 • II 8.70 • III 8.10 • IV 7.89 • V 8.09 • VI 8.67 → CGPA 8.22

Coursework: Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks, Software Engineering, Object Oriented Programming

Publication: "Digital Heritage Preservation Technologies for Monasteries of Sikkim", Advancement in Image Processing and Pattern Recognition, 2026.

POSITIONING LINES — PICK ONE PER APPLICATION

Software Engineer — Information Science Engineering student graduating May 2027 who ships systems end to end — REST and WebSocket backends, React and Next.js front ends, Python inference pipelines. Maintains a live public site gated by a five-check audit suite, and built a real-time assistant with three-provider LLM failover and measured latency. Hackathon winner and published author in image processing.

AI / ML — Information Science Engineering student graduating May 2027 training and shipping computer-vision and multimodal AI systems in Python — CNN classification on clinical imaging with a calibrated abstention gate, YOLO-based detection on drone imagery, and LLM agents with multi-provider routing and latency telemetry. Published author in image processing and first-place agentic-AI hackathon winner.

Data Analytics — Information Science Engineering student graduating May 2027 working across the full analytics stack — SQL and Python for cleaning and exploratory analysis, Tableau and Power BI for decision-facing dashboards, and machine learning where descriptive analysis runs out. Spent a full internship on market sizing, competitive analysis and unit economics across two international markets.

Product & Strategy — Information Science Engineering student graduating May 2027 who defines product direction for technical systems and can hold the conversation with the engineers building them. Founding member of TerraHawk, where I own requirements, scope boundaries and the automation-versus-human-review threshold; spent a full internship on market, competitive and unit-economics analysis across two international markets.

QA / Quality Engineering — Information Science Engineering student graduating May 2027 applying for QA because I already do the work informally and want to learn it properly. On my live public project I wrote five automated audit gates that run before every release, and one caught a contrast check passing on text nobody could read. Comfortable with JavaScript, TypeScript, Python, SQL and REST APIs; keen to learn test design, automation and release process alongside an engineering team.

COMPLETE SKILLS INVENTORY

Languages: Java, JavaScript, Python, SQL, TypeScript

Backend: asyncio, Express, FastAPI, Node.js, REST API design, server-side validation, WebSockets

Frontend: CSS3, HTML5, Next.js 16, React 19, responsive and accessible UI, Tailwind CSS

Databases: indexing, joins, migrations, MongoDB, MySQL, PostgreSQL, query optimisation, SQLite — schema design

AI / ML: Keras, LangChain, LLM integration, OpenCV, PyTorch, scikit-learn, TensorFlow, YOLOv5

Fundamentals: Computer Networks, Data Structures and Algorithms, DBMS, OOP, Operating Systems, System Design

Tools: Agile, DevTools, Git, GitHub, headless browser automation, Jupyter, Lighthouse, Linux, Postman, REST APIs, VS Code

Programming: aggregations, Java, JavaScript, JavaScript / TypeScript, Matplotlib, NumPy, Python, Python (pandas, SQL, SQL (joins, window functions)

Machine Learning: Class-Imbalance Handling, Classification, Clustering, CNNs, Hyperparameter Tuning, Regression, Transfer Learning

Computer Vision: augmentation, facial and posture analysis, Feature Pyramid Networks, OpenCV, preprocessing, YOLOv5 object detection

Generative AI: agentic workflows, Gemini, LangChain, LLM integration and multi-provider routing (Groq, OpenAI), prompt engineering

Frameworks: Keras, Matplotlib, NumPy, OpenCV, pandas, PyTorch, scikit-learn, TensorFlow

Evaluation: Balanced Accuracy, Confusion Matrix, cross-validation, dataset auditing, entropy-based abstention, F1, Precision, Recall

Serving and Data: asyncio, FastAPI, Git, Linux, MongoDB, PostgreSQL, REST APIs, SQLite, WebSockets

Analytics: Cohort and Trend Analysis, Dashboarding, Data Cleaning, Exploratory Data Analysis, Feature Engineering

BI and Reporting: Excel (pivots, lookups, modelling), Power BI, Tableau

Statistics and ML: Balanced Accuracy, class-imbalance correction, Classification, Clustering, Cross-Validation), F1, model evaluation (Precision, Recall, Regression

Business: Competitive Analysis, Market Research, Market Sizing, Operations Modelling, Unit Economics

Product: acceptance criteria, beachhead selection, PRD and BRD authoring, prioritisation, Requirements definition, scope and non-goals, workflow design

Strategy: competitive analysis, Market research, market sizing, operating models, regulatory context mapping, unit economics

Data and BI: dashboarding, Excel, exploratory data analysis, metric definition, Power BI, SQL, Tableau

Technical Fluency: Java, JavaScript / TypeScript, MongoDB, Next.js, Node.js, PostgreSQL, Python, React, REST APIs

AI / ML Literacy: Computer vision, confidence thresholds and model evaluation trade-offs, LangChain, LLMs, prompt engineering

Collaboration: Agile, async cross-timezone working, Stakeholder communication, technical documentation

Testing and Quality: defect investigation and root cause analysis, edge case and boundary analysis, Manual and exploratory testing, regression checks, REST API testing, test case design

Automation and Audit: axe-core accessibility auditing, Core Web Vitals, Headless browser automation, Lighthouse, responsive testing, scripted pre-release gates, WCAG contrast checking

Web and Backend: FastAPI, HTTP status codes, JSON, Next.js, Node.js, React, REST APIs, WebSockets

Databases and Practice: Agile, documentation, Git, GitHub, indexing, Linux, migrations, SDLC, SQL queries and joins, SQLite schema design, structured logging

PROJECTS — EVERY BULLET, GROUPED BY PROJECT

TerraHawk

Founding member, three-person venture | preparing to apply for incubation | Python · SQL · PostgreSQL · Decision Thresholds | Founding member, three-person venture | Python · YOLO · Feature Pyramid Network · OpenCV | Founding member, three-person venture | React · Node.js · Postgre

- Architecting a full-stack platform that verifies reported agricultural activity actually occurred, issuing a tamper-evident Truth Ticket: claim, evidence, model output, confidence score, chain of custody.
- Designed the evidence-fusion layer merging drone detections with satellite and field data into one confidence-scored record — routine claims clear automatically, low-confidence cases route to human review.
- Specified REST contracts between detection service, evidence store and downstream insurance and lending consumers.
- Specified a YOLO detector with a Feature Pyramid Network for multi-scale crop-disease and field-condition detection on drone imagery.
- Instrumented training with dataset audit checks and configurable augmentation presets, surfacing label and distribution defects before training runs rather than after.
- Designed multi-source evidence fusion — detections, geospatial signals, temporal imagery, metadata, human review — into one confidence-scored decision with an explicit escalation threshold.
- Designed the logic converting model output plus satellite and field evidence into one confidence-scored record, and set the thresholds separating auto-cleared claims from escalations — the trade-off driving operating cost and auditability.
- Defined the Trust Score metric and the evidence record consumed by insurance claim assessment and agricultural lending workflows.
- Built dataset audit checks into the pipeline to catch label and distribution defects before they propagate into reported numbers.
- Own product direction and requirements for a platform verifying whether reported environmental and agricultural activity took place, issuing a tamper-evident Truth Ticket: claim, evidence, model output, confidence, chain of custody.
- Scoped the core workflow trade-off — high-confidence claims clear automatically, ambiguous ones route to human review — which drives operating cost, auditability and trust with regulated buyers.
- Identified insurance claim assessment and agricultural lending as the two beachhead use cases and defined the evidence each demands.
- Drove the product through repeated feasibility rewrites: a weed-detection concept became a verification platform because each iteration was tested against how assessors and lenders actually work. Documented explicit non-goals to keep the build boundary clear.
- Specified risk-based triage: low-risk cases clear automatically, uncertain or anomalous ones escalate — the same trade-off as deciding what a suite can pass unattended and what a person must review.
- Defined the Truth Ticket record so every decision the platform makes can be reconstructed afterwards.

Riches Garden

Next.js 16 · React 19 · TypeScript | live in production, public repository | Next.js 16 · React 19 · TypeScript · axe-core | live in production, public repository | Next.js · TypeScript | live in production | Next.js · TypeScript · Core Web Vitals · axe-core

- Ship and maintain a live site used by a residents' committee, gated by a five-check pre-release suite: link and asset integrity, axe-core accessibility, contrast across seven viewports, and Core Web Vitals page weight.
- Cut Kannada webfont payload 89 kB → 14 kB by subsetting to the 33 glyphs in use, then fixed the conjunct-cluster corruption naive subsetting introduced.
- Built a five-check suite reporting link integrity, accessibility violations, contrast ratios across seven viewports and page-weight metrics before every release, turning subjective judgement into measured pass/fail data.
- Ship and maintain a live site a residents' committee uses for their annual festival, gated by an automated pre-release quality suite so the yearly content rollover cannot regress the experience.
- Wrote a five-gate suite running before every release: link and asset integrity with console-error capture; axe-core accessibility with heading-order and landmark checks; contrast, overflow and tap-target size across seven viewports; page weight with Core Web Vitals.
- Found a false pass in the contrast check — it read declared CSS colours, but the text sat over a background image, so it reported a value that never reached the screen. Rewrote it to sample rendered pixels.
- Caught a silent rendering defect: subsetting Kannada webfonts to the 33 glyphs in use cut 89 kB → 14 kB, but broke conjunct clusters because the shaper decomposes vowel signs before substitution — no error, no warning, only wrong glyphs.

AI Lifestyle Mirror

Python · OpenCV · asyncio · WebSockets · Groq / Gemini / OpenAI | Python · asyncio · WebSockets · SQLite | Python · asyncio · WebSockets · SQLite · OpenCV · Groq / Gemini / OpenAI

- Built a presence-triggered smart mirror serving time, weather, news and camera-based posture coaching, with next-day recall from local history.
- Routed LLM calls through three-provider failover ordered by latency and cost behind 8- and 10-second timeouts, logging per-provider latency to a diagnostics endpoint.
- Integrated four REST services behind one orchestration layer — event bus, state manager, WebSocket broadcaster, and a cache that flags stale entries rather than serving them silently.
- Ran concurrent inference streams against a live camera feed — facial analysis, emotion recognition, posture monitoring — fused with LLM assistance and live news and weather APIs.
- Designed a three-provider LLM router ordered by latency and cost with per-call timeouts and automatic failover, recording per-provider latency to a diagnostics service.
- Built failure paths before the happy path: three-provider failover behind timeouts with latency logging, a cache that flags stale entries rather than serving them silently, and a finally-path returning the state machine to idle after a crash.

DermaCare

PyTorch · MobileNetV2 · FastAPI · Next.js | Python · pandas · PyTorch · scikit-learn

- Trained a MobileNetV2 classifier over the 10,015-image HAM10000 set (7 classes, class-weighted loss), served through FastAPI with a Next.js front end.
- Added a dual abstention gate — declines below 0.15 confidence or above 1.85 entropy — evaluated on balanced accuracy.
- Trained a MobileNetV2 transfer-learning classifier over 10,015 dermatoscopic images across 7 lesion classes, class-weighting the loss to correct severe imbalance.
- Audited the inference path and found a temperature divisor sharpening the output distribution without improving the prediction — inflating displayed confidence and weakening the abstention gate beneath it.
- Built a dual abstention gate (below 0.15 confidence or above 1.85 entropy) and evaluated on balanced accuracy, precision, recall and F1, since false negatives carry the clinical cost.

- Profiled 10,015 images across 7 classes, quantified severe imbalance and corrected it through class-weighted loss and stratified splits.
- Chose metrics around cost asymmetry — balanced accuracy, precision, recall, F1 over raw accuracy — and added an entropy-based abstention rule for low-certainty cases.
- Audited the inference path and found a defect only the user would see: a temperature divisor sharpened the output distribution without improving the prediction, inflating on-screen confidence and weakening the abstention gate beneath it.
- Added a dual abstention gate (below 0.15 confidence or above 1.85 entropy) and evaluated on balanced accuracy, since raw accuracy hides minority-class failure in the 10,015-image HAM10000 set.

Veniteck

Market sizing · Competitive benchmarking · Operating models · Excel

- Sized the addressable market and modelled operating economics for an on-demand driver service targeting Australia, producing the analysis behind go-to-market and pricing decisions.
- Benchmarked competitors and mapped regulatory context for an agri-tech initiative entering Papua New Guinea.

Veniteck Solutions

Market sizing · Competitive analysis · Unit economics · Operating models

- Produced the market opportunity and operating-model analysis behind go-to-market and unit-economics decisions for an Australia-focused mobility venture.
- Ran competitive analysis and business development support, including regulatory context, for an agri-tech initiative entering Papua New Guinea.

Digital Heritage Preservation

AI · VR | Published research, 2026

- Scoped and built an AI and VR system for monastery digitisation and virtual tourism, published in Advancement in Image Processing and Pattern Recognition (2026).

EXPERIENCE

Business Strategy Intern — Veniteck Solutions

May 2026 – August 2026

Bangalore, India

- Turned open-ended briefs for two international ventures — an on-demand driver service in Australia and an agri-tech initiative in Papua New Guinea — into structured written requirements and evaluated options.
- Coordinated asynchronously across time zones, clarifying requirements in writing and documenting findings clearly enough to stand without a follow-up call.
- Evaluated market opportunity, operating models and unit economics for DrvMee, an Australia-focused mobility venture, producing the analysis behind go-to-market decisions.
- Ran competitive analysis and supported business development and strategic planning for OPMS, an agri-tech initiative entering the Papua New Guinea market.
- Delivered market and operating-model analysis across two distinct international markets and their regulatory contexts, working from open-ended briefs to written deliverables.

ACHIEVEMENTS & LEADERSHIP

- 1st Place, Cicada Agentic AI Hackathon — designed, built and deployed a working agentic AI system inside the event window. • Winner, Machine Learning Expo and Generative AI Expo.
- Smart India Hackathon 2025, National round (SIH25061) — led system architecture and technical documentation for a government-issued problem statement. • Event Organiser, IEEE Student Chapter — 5+ technical events, 100+ attendees.

CERTIFICATIONS

ML and Deep Learning Specialization (Stanford Online / DeepLearning.AI) • Software Engineering, Cloud Computing and Cyber Security (Infosys Springboard) • Full Stack Development (SimpliLearn) • Java (IIT Bombay)

Languages: English, Hindi, Kannada, Malayalam, Tamil

CLAIMS FROM OLDER DRAFTS — VERIFY BEFORE REUSING

These appeared in 2026 drafts but are not in the current variants. Add one back only if you can walk an interviewer through it: Docker deployment; JWT authentication; Prisma ORM; Socket.IO; microservices; A/B testing; 4K at 90 FPS VR performance benchmark for the Sikkim system; stakeholder mapping and negotiation as named skills.